Examiner only

4. The following table shows the composition of the atmosphere.

Gas	Chemical formula	Percentage found in the atmosphere (%)
argon	Ar	0.93
carbon dioxide	CO_2	0.0360
helium	He	0.0005
hydrogen	H_2	0.00005
methane	CH_4	0.00017
neon	Ne	0.0018
nitrogen	N_2	78.08
nitrous oxide	N_2O	0.00003
oxygen	O_2	20.95
ozone	O_3	0.000004

Use the table to answer parts *(a)* and *(b)*.

(a) (i) Name **two** gases that occur as single atoms. [1]

....

(ii) Name **two** elements that occur as molecules. [1]

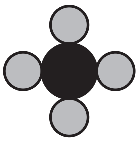
....

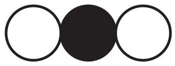
(iii) Name the gas that has the **lowest** percentage. [1]

....

Examiner only

(b) (i) If  represents a carbon atom, name the gases from the table that can be represented by the following diagrams. [2]





....

(ii) Use the information in part (i) to draw the diagram for oxygen gas. [1]

(c) The atmosphere can become polluted by substances known as chlorofluorocarbons (CFCs). One example of a CFC is CH_2ClF.

3410U101 09

(i) State how many hydrogen atoms are present in CH_2ClF. [1]

....

(ii) Give the **total** number of atoms shown in the formula, CH_2ClF. [1]

....

8